

Research Assignment (Project Mentor: Monisha)

Personality-Aware Engagement Prediction in Online Learning

Reference Paper:

Title: *Personality-Aware Engagement Prediction in Online Learning*

Link: <https://dl.acm.org/doi/pdf/10.1145/3746270.3760234>

Dataset Access:

Obtain access to the dataset by signing the EULA:

<https://sites.google.com/view/mocse-dataset/download?authuser=0>

NOTE: You will work on a subset of the dataset. Make sure your subset has a balanced representation of all the labels in the dataset.

Task 1: Feature Extraction [15 Marks]

Explore multimodal feature representations for engagement prediction.

You may use handcrafted features, pretrained model features, or a combination of both.

Examples of Feature Choices (Not Mandatory)

- Video (handcrafted): OpenFace features such as eye gaze, head pose, and facial Action Units (AUs)
- Audio (handcrafted): OpenSMILE features (e.g., prosodic, spectral, and energy-based descriptors)
- Pretrained features (optional): Vision, audio, or multimodal embeddings from pretrained models

Note: OpenFace and OpenSMILE are provided as examples only. Students are free to use other handcrafted features or pretrained representations, provided the choices are well-motivated.

Task 2: Multimodal Feature-Based Engagement Modeling [20 Marks]

Explore and evaluate the suitability of several classical machine learning models for prediction using multimodal behavioral features. As the extracted features are temporal sequences, this task will investigate methods for converting sequential data into fixed-length representations compatible with traditional ML algorithms. Candidate models include **Logistic Regression** (interpretable baseline), **Support Vector Machines (SVM)** (effective for high-dimensional structured features), **Random Forests** (robust to noise and nonlinear relationships), **Gradient Boosting methods such as XGBoost/LightGBM** (strong performance on tabular behavioral data), and **Multilayer Perceptrons (MLP)** (to capture nonlinear feature interactions).

To incorporate temporal information, the following transformation strategies can be explored:

- Temporal aggregation: summarize each time window using statistical descriptors (e.g., mean, variance, extrema) to capture overall behavioral patterns.
- Dynamic feature engineering: encode temporal change through derivatives, trends, or motion-based features reflecting behavioral dynamics.
- Sliding-window representation: treat short overlapping temporal segments as independent samples to preserve local temporal context.
- Sequence flattening: concatenate time steps into a single feature vector to retain ordering information in a fixed-size input.
- Temporal pooling: apply pooling operations (e.g., mean/max pooling) across time to emphasize dominant or salient behavioral signals.

The effectiveness of each representation will be evaluated in terms of prediction performance, sensitivity to temporal changes, and generalization across unseen data and participants. Report your observations.

Task 3: Personality-Conditioned Temporal Modeling [15 Marks]

In this task, you will design a long-range temporal engagement prediction model that explicitly accounts for annotation noise and missing temporal segments, and analyze how these issues affect engagement prediction across different personality traits.

Unlike Task 2, which focuses on multimodal interactions within short temporal windows, this task emphasizes robust temporal reasoning and error analysis.

What You Need to Do:

1. Build a novel long-range temporal model (e.g., temporal transformer, memory-based model, or hierarchical temporal architecture) that uses:
 - Multimodal features (from Task 2)
 - Personality traits
 - Engagement predictions over extended temporal sequences
2. Design the model or training strategy to be robust to noisy labels and missing chunks, for example through:
 - Temporal smoothing or consistency constraints
 - Masked or variable-length sequence modeling
 - Loss functions
3. Predict engagement at future time steps or over long temporal horizons.

Evaluation:

- Compare performance with short-context multimodal models from Tasks 2
- Report standard engagement metrics (e.g., CCC, F1, MAE)
- Evaluate generalization across participants or personality groups